

# **ALTERNATIVES TO THE ANISOTROPIC STENCIL FILTER**

Elliptical & Rectangular Gaussian Designs

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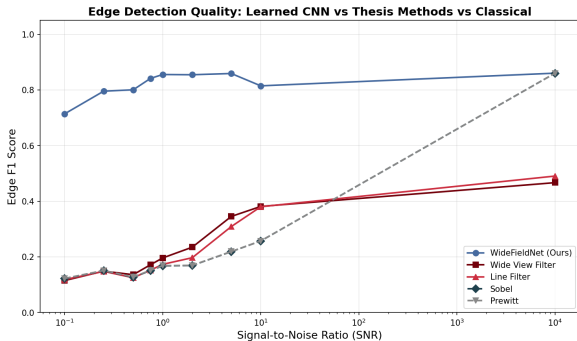
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# MOTIVATION

From WideFieldNet to filter critique to  
geometric kernels

# IT STARTED WITH A CNN

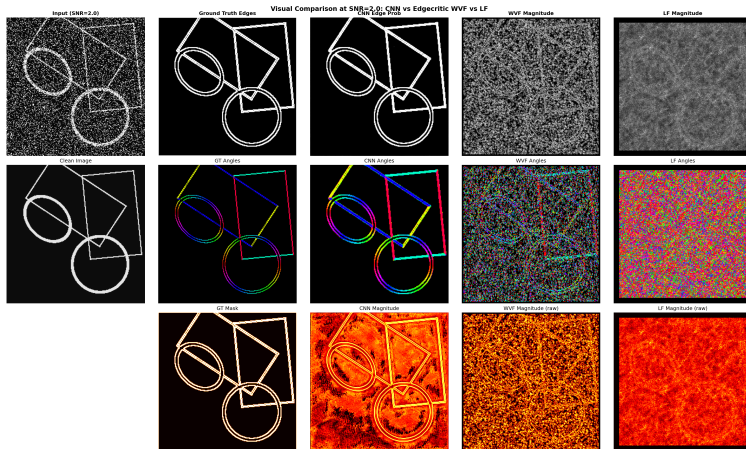
Developed WideFieldNet, a dilated U-Net (2.6M params) for edge detection on synthetic data with controlled noise. Results against Bagan's WVF/LF were unexpected.



WideFieldNet dominated across all SNR levels on synthetic data.

But the WVF/LF scores were far below what Bagan reported. This prompted a deeper investigation into the filter parameters.

# CNN VS WVF/LF ON SYNTHETIC DATA





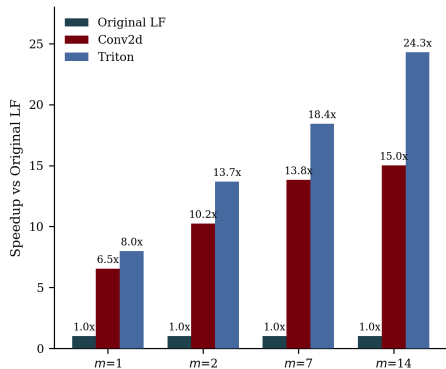
# FOUR REPORTS OF INVESTIGATION

The CNN results led to a systematic critique of the WVF/LF filters.

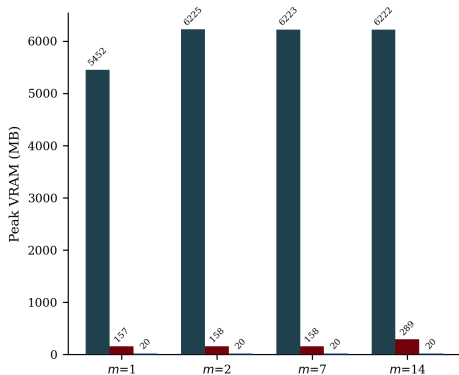
1. Parameter ablation — Bagan's published parameters ( $N_p=250$ ,  $N_s=18$ ,  $d=4$ ) are suboptimal. Optimal:  $N_p=25$ ,  $N_s=4$ ,  $d=2$ . ODS gains of +0.01 to +0.16.
2. Benchmarking — Optimized WVF beats all 54 classical operators. DL models are 0.09 ODS higher but  $400\times$  slower.
3. Noise robustness — Bagan's large  $N_p$  was optimal for noisy underwater data. Optimal  $N_p$  scales inversely with SNR ( $25\rightarrow 400$ ).
4. Statistical validation — Nonparametric tests confirm all claims. BSDS500 is an outlier dataset ( $\rho < 0.17$  with others).

This also led to developing a GPU-accelerated ASF reformulation — collapsing the LF pipeline into a single fused convolution ( $24\times$  speedup,  $311\times$  VRAM reduction).

# ASF SPEEDUP RESULTS



(A) Speedup vs Original LF



(B) Peak VRAM (MB)

24 $\times$  wall-clock speedup at  $m=14$ , 311 $\times$  VRAM reduction (6.2 GB  $\rightarrow$  20 MB). The ASF reformulation

# BUT THE FUSED WEIGHTS ARE WASTEFUL

Examining the fused stencil weights revealed a deeper issue.

The ASF is faster than the naive LF, but the stencil it produces is irregular. Many pixels carry near-zero or cancelled weight due to overlapping neighborhood contributions.

## Key question

The polynomial fitting, pseudoinverse, and deduplication pipeline produces a weighted kernel. Can we define the kernel shape directly and assign weights geometrically, skipping the polynomial machinery entirely?

# THE OVERLAP PROBLEM

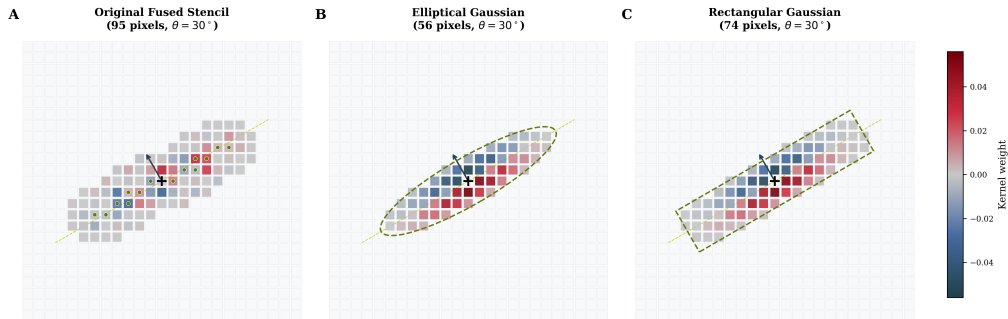
When virtual pixel neighborhoods overlap and their contributions are fused, many pixels receive near-zero or cancelled weights.

## Observation

At  $m=7$ ,  $N_p=20$ : the stencil uses 91 pixels, but most carry negligible weight. The polynomial fit and deduplication produce irregular, unpredictable weight distributions.

Can we define the kernel shape directly and assign weights geometrically?

# KERNEL WEIGHT COMPARISON



A: Fused stencil (95 px, irregular, scattered outliers). B: Elliptical Gaussian (56 px, smooth). C: Rectangular Gaussian (74 px, clean boundary). All matched to the same effective aspect ratio.

# GEOMETRIC KERNEL DESIGNS

Elliptical and rectangular Gaussian  
kernels

# TWO GEOMETRIC ALTERNATIVES

Both designs share the same core structure. Rotate pixel coordinates into the kernel frame, apply a distribution, and multiply by the derivative profile.

## Elliptical Gaussian

Boundary:  $\frac{u^2}{\sigma_a^2} + \frac{v^2}{\sigma_c^2} \leq 9$

Weight:  $K = -v \cdot \exp\left(-\frac{u^2}{2\sigma_a^2} - \frac{v^2}{2\sigma_c^2}\right)$

- Smooth falloff at boundary
- Fewest active pixels
- Best at high noise

## Rectangular Gaussian

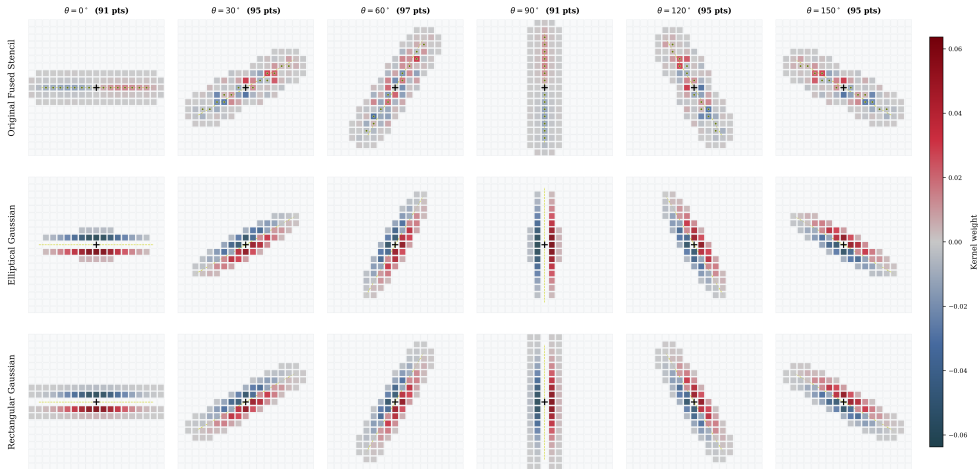
Boundary:  $|u| \leq h_a, |v| \leq h_c$

Weight: same Gaussian  $\times (-v)$ , clipped to rectangle

- Hard boundary, Gaussian inside
- More active pixels than ellipse
- Slight edge on clean images

Every pixel gets exactly one weight, determined solely by its geometric position. No polynomial fitting, no pseudoinverse, no overlapping neighborhoods, no cancellation.

# STENCIL WEIGHTS BY ORIENTATION





# PERFORMANCE

3× faster, same accuracy

# SPEED COMPARISON

All three filters at matched parameters ( $\sigma=2.0$ , 36 orientations,  $640 \times 427$  image). Median of 10 runs.

| Filter                   | Time (ms) | Speedup | ODS F-score |
|--------------------------|-----------|---------|-------------|
| Original (fused stencil) | 4,013     | 1.0×    | 0.4221      |
| Elliptical Gaussian      | 1,201     | 3.3×    | 0.4211      |
| Rectangular Gaussian     | 1,424     | 2.8×    | 0.4220      |

- ODS scores within 0.3% across all three designs
- Elliptical is fastest due to fewest active pixels (compact elliptical mask)
- No polynomial fitting, no pseudoinverse, no deduplication step

# SIDE-BY-SIDE FILTER OUTPUT



# EDGE DETECTION RESULTS

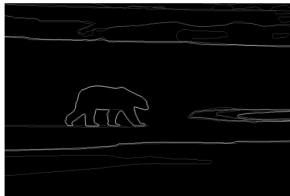
BSDS500 test set, 10 images

# CLEAN IMAGE COMPARISON

Input



Ground Truth



ODS F-scores:

Original: 0.608  
Elliptical: 0.605  
Rectangular: 0.605

Original (F=0.608)



Elliptical (F=0.605)



Rectangular (F=0.605)



# CLEAN IMAGE COMPARISON (CONT.)

Input



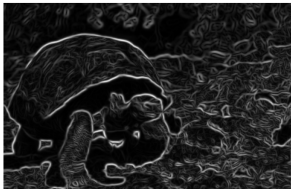
Ground Truth



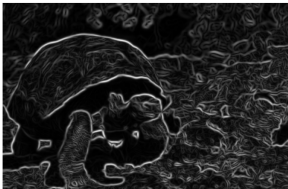
ODS F-scores:

Original: 0.381  
Elliptical: 0.378  
Rectangular: 0.380

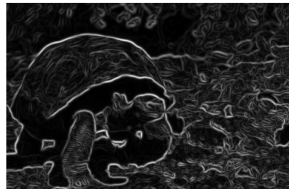
Original (F=0.381)



Elliptical (F=0.378)



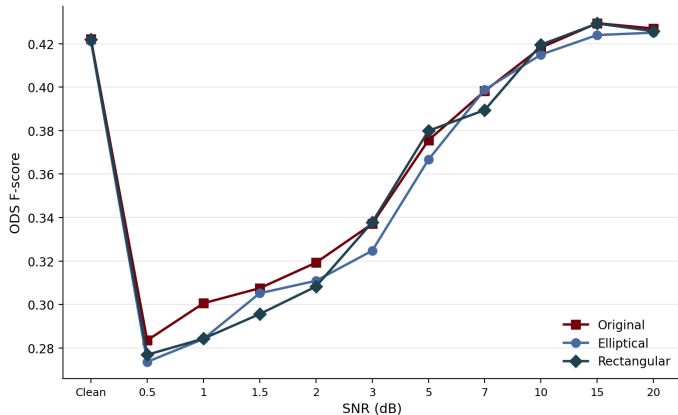
Rectangular (F=0.380)



# **NOISE ROBUSTNESS**

Does the polynomial fit help in noise?

# SNR VS ODS



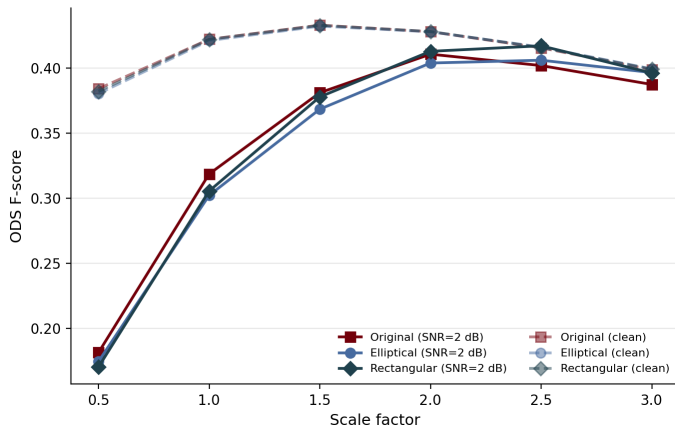
Original holds a 1–2% ODS edge at extreme noise (SNR < 3 dB).

Above SNR 10 dB, all three converge.

Is this polynomial fitting, or just kernel size?



# SCALING UP CLOSES THE GAP



Solid: SNR=2 dB.

Dashed: Clean.

The noise advantage disappears at scale 2.0x.

Noise robustness is a function of kernel size, not polynomial order.

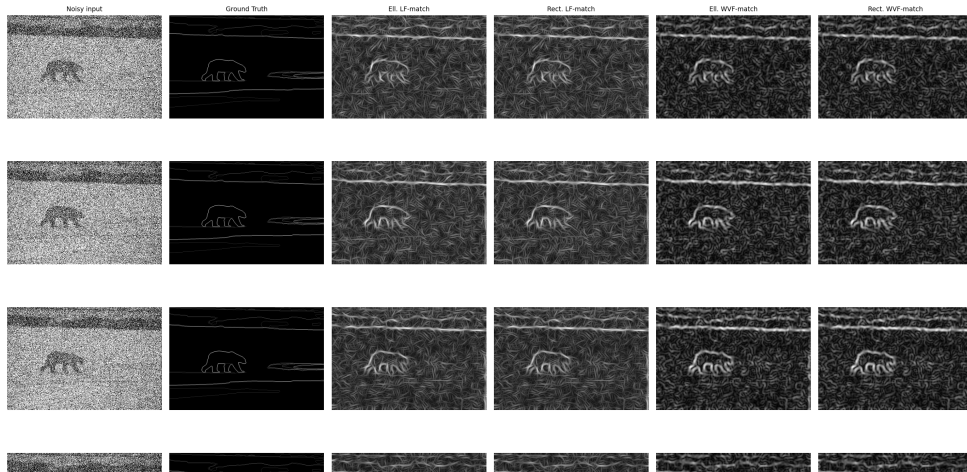
# NOISE ABLATION: PARAMETERS SHIFT

The ASF ablation revealed that optimal parameters change dramatically with noise level.

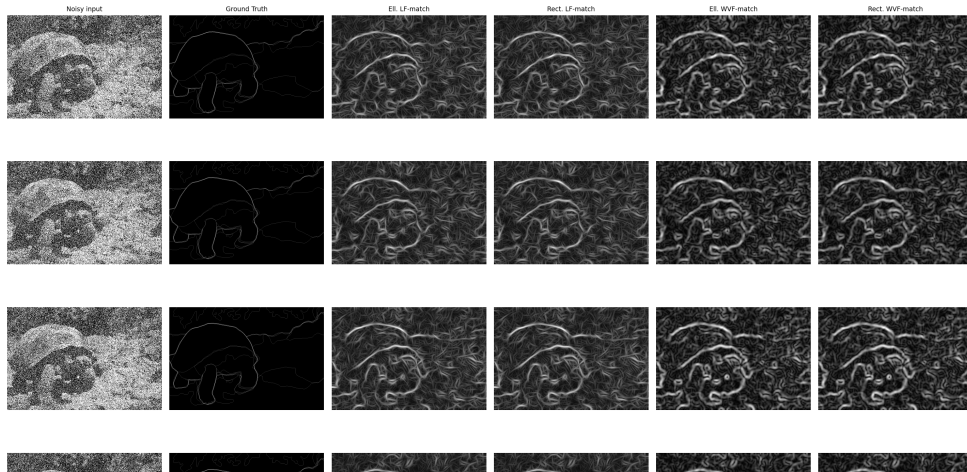
| Condition        | Best $N_p$ | Best $N_s$ | Best $m$ | Trend                          |
|------------------|------------|------------|----------|--------------------------------|
| Clean            | 25–40      | 3–4        | —        | Small kernel, few orientations |
| Gaussian SNR=4   | 65         | 3          | 5        | Larger kernel                  |
| Gaussian SNR=2   | 100        | 3          | 7        | Even larger                    |
| Gaussian SNR=1   | 300        | 72         | 10       | Much larger, many orientations |
| Gaussian SNR=0.5 | 400        | 72         | 14       | Maximum size                   |
| Gaussian SNR=0.3 | 400        | 36         | 14–20    | Saturated                      |

At extreme noise, the LF with  $m=14$  uses massive elongated stencils ( $43 \times 15$  pixels). The geometric kernels replicate this with  $\sigma_a=7.17$ ,  $\sigma_c=2.50$ .

# EDGE MAPS UNDER HEAVY NOISE



# EDGE MAPS UNDER HEAVY NOISE (CONT.)

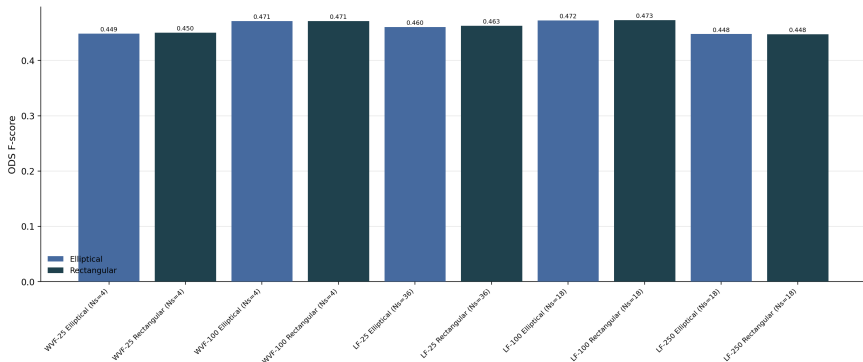


# **ABLATION-MATCHED RESULTS**

Matched to ablation-optimal ASF  
configurations

# MATCHED TO OPTIMAL ASF SETTINGS

The ASF ablation identified optimal configurations per dataset. We matched elliptical and rectangular kernels to each by measuring the fused stencil's pixel extent and deriving equivalent  $\sigma$  values.



# SUMMARY

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The fused stencil's polynomial machinery is unnecessary for edge detection:

- Overlapping virtual-pixel neighborhoods cancel weights and waste pixels
- 91–97 active pixels, many carrying near-zero weight
- Requires pseudoinverse, polynomial fitting, and deduplication

Geometric kernels achieve the same accuracy with a simpler design:

- Elliptical Gaussian: fewest pixels, cleanest weights, best under noise
- Rectangular Gaussian: slightly more pixels, marginally better on clean images
- Both are  $3\times$  faster at matched parameters, with  $<0.3\%$  ODS difference

Noise robustness is a function of kernel size, not polynomial order:

- The original's small noise advantage vanishes when geometric kernels are scaled up
- Ablation confirms: heavy noise demands larger kernels and more orientations, not higher-order polynomial fits



# THANK YOU!

Questions & Discussion

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